Curriculum Vitae

Chung Shing Rex Ha — Biostatistician

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Permanent residence in Germany (*Niederlassungserlaubnis*) — unrestricted right to work, no employer sponsorship required.

Experience

Research Associate — IMBEI, Universitätsmedizin Mainz

Oct 2019 – Nov 2025 · Department of Genomic Statistics and Bioinformatics

- Study statistician for the quantitative evaluation of a 2,451-patient non-randomised controlled trial — sole author of the statistical analysis plan; performed all primary and secondary endpoint analyses.
- Modelled recurrent healthcare-utilisation endpoints from routine statutory-insurance data using zero-inflated negative binomial regression with 1:2 covariate matching, reporting incidence-rate ratios.
- Co-authored the formal evaluation report to the German Federal Joint Committee (G-BA Innovationsausschuss) for a publicly funded trial, owning the quantitative evaluation.
- Integrated electronic patient records from a regional physicians' association and six statutory health insurers into analysis-ready datasets, with cross-source quality control and conflict reconciliation, together with our data-management team.
- Developed a data management plan and data transfer specification with a data-management specialist for external health-insurance data providers, plus a data review plan comparing internal against each external source.
- Built machine-learning pipelines and interactive dashboards for pancreatic-cancer biomarker discovery on next-generation proteomics and LC-MS metabolomics.
- Restructured a single-table retrospective dataset into normalised relational tables suitable for analysis and publication.
- Provided statistical consulting to medical doctoral candidates and research groups across more than fifteen clinical departments.

Research Associate — Institute of Genetic Epidemiology, Helmholtz Zentrum München

Jun 2019 – Sep 2019

- Constructed detailed pedigrees for complex family structures to improve genomic analysis of familial pancreatic cancer.

Research Assistant — Institute of Genetic Epidemiology, Helmholtz Zentrum München

Sep 2018 – May 2019

- Applied machine learning to metabolic differences between cancer patients and healthy individuals for early detection.

Junior Market Analyst — Autoduck Industrial Ltd, Hong Kong

Aug 2013 – Oct 2014

- Monthly sales-activity summaries to evaluate marketing strategy; market-trend analysis to identify prospective customers.

Education

PhD, Biology — Johannes Gutenberg University Mainz · 2023 – present

Omics-based Biomarker Discovery for Familial Pancreatic Cancer using Machine Learning. Write-up stage; submission targeted end 2026 / early 2027.

M.Sc. Biostatistics — Ludwig Maximilian University of Munich · 2019

Thesis: *Bayesian Blind Estimation of the Arterial Input Function in Dynamic Contrast-Enhanced MRI* (R, Stan).

B.Sc. Investment Science — Hong Kong Polytechnic University · 2013

Higher Diploma in Financial Planning — Hong Kong

DSH-2 — Ludwig Maximilian University of Munich · 2016
German language qualification for university admission.

Selected projects

FAMOUS — case-based care of multimorbid patients by advanced practice nurses. Innovationsfonds / G-BA, FKZ 01NVF19012. Two-arm non-randomised multicentre interventional study, 2,451 patients. Sole author of an estimand-based statistical analysis plan (ICH E9(R1)) defining a primary and four secondary estimands; Bayesian zero-inflated negative binomial primary analysis in `brms`/Stan.

FaPaCa — German National Case Collection for Familial Pancreatic Cancer. Data management and statistical analysis; machine-learning biomarker pipelines on next-generation proteomics and LC-MS metabolomics. Basis of the doctoral work.

IMPETUS — randomised controlled trial in elective colorectal surgery. A priori power calculation, contribution to the ethics-committee submission, and the primary-endpoint analysis.

Methods

- Survival analysis (Cox regression, Kaplan–Meier, log-rank), mixed models, generalised estimating equations, bootstrap.
- Bayesian modelling in R with Stan and `brms` — priors, MCMC diagnostics, posterior summaries and credible intervals.
- Sample-size and power calculations across clinical trials, observational studies and animal-research applications, including Bayesian precision-based determination and recurrent-event planning with intra-patient correlation and cluster effects.
- Model-based gradient boosting, stability selection, Bayesian additive regression trees and Gaussian graphical models on high-dimensional omics data.
- Explainable ML — SHAP, LIME, partial dependence plots, stability selection.
- Propensity and covariate matching (`MatchIt`).

Programming and tools

R (Shiny, pharmaverse) · Stan · Python · PostgreSQL · bash · Git · Docker · Snakemake · Linux

Familiar with CDISC concepts (SDTM, ADaM); building hands-on ADaM practice with the R pharmaverse.

Teaching and supervision

- Taught regression analysis in the cross-disciplinary Q1 module for medical students, IMBEI · 2020–2024.
- Ran hypothesis-testing workshops for doctoral researchers at the Center for Thrombosis and Hemostasis, Universitätsmedizin Mainz · 2023–2024.
- Supervised bachelor's and master's students and research assistants; co-supervised a bachelor thesis.

Award

1st place, 2nd curATime Hackathon "curAHack" — Gutenberg Digital Hub Mainz, 2025. Multiclass classification of anti-CRISPR proteins.

Languages

German (fluent, negotiation level) · English (fluent, negotiation level) · Cantonese (native) · Mandarin (good working knowledge)

Publications

See the publications page.